

Four Types of Connections Used in Light Steel Framing NZ

Using light steel framing in construction can be a tricky task, but the good news is that there are various way of completing that task. In general terms there are four different types of connections used with light steel framing and they are welded connections, bolted connections, pressjoining/clinching connections and riveted connections.

Welded

Although welded connections create a strong join which is up to the task of handling the stress placed on them, it is a method which does present a couple of problems.

The main source of these problems is the fact that the welding itself has to take place on the construction site. This means that external factors such as the wind causing the steel members to shift can lead to the connection created being less than 100% perfect.

Press Joining/Clinching

Press joining, which is also known as clinching, is a process for joining light steel framing which has been compared to a puzzle – although it is a puzzle which is capable of creating a strong join. It is a method which involves a combination of drawing and forming which is used to lock two or more layers of steel in place together.

It is undoubtedly an effective means of connecting light steel framing, but it is also rather time consuming, which makes it the least popular of the methods of connection.

Riveted

Riveted connections come in two different types – self-piercing rivets and blind rivets. Blind rivets require pre-drilled holes to be in place, unlike self-piercing rivets, but despite the differences between the two, they are both capable of creating a strong, durable join with an impressive pull-out resistance.

The drawback to rivets is that they perform best when used with thin connections such as the seams along profiled sheeting and decking. In addition to this, self-piercing rivets involve the use of a hydraulic operated tool and a process which has multiple steps. These are the factors which mean that, though they may be extremely useful, rivets can't really compare with the option of bolted connections.

Thin Wall Bolts and Wall Bolts

The fact that wall bolts are the most common method of connecting lighting steel sections is, in large part, down to how simple they are to use. This simplicity, however, doesn't make them any less effective when in place. Standard wall bolts come into their own during the roll forming process, thanks to the fact that they can easily punch through the steel sectioning.

Although the strength and capacity of a connection between bolt and steel is usually a reflection of the strength of the thinner material involved, wall bolted connections are still durable and effective. They make the work of light steel framing a quick and simple task.

The thin wall bolt – also known as a TW bolt – was developed by BlindBolt to provide a blind fixing for use with cladding material or lightweight steel sheet.

TW bolts offer the unbeatable combination of quality and convenience. The fact that the installation of the TW bolts is so quick and simple helps to speed up any assembly work, and saving time means saving money. In addition to this the finished join is both durable and visually appealing, thanks to the low visible profile of a TW bolt.

When compared with the other types of connection thin wall bolts offer the advantages of being easy to use while creating a high quality durable join which is also aesthetically pleasing. Put all of that together and you have the ultimate connection for light steel framing.

Bolted Shut

The experts at BlindBolt created thin wall bolts (TW bolts) to provide a blind fixing for use with lightweight steel sheets and cladding materials. In the time since we have constantly strived to further develop the concept of thin wall bolts, making sure that we can offer our customers the best possible blind fixing solutions.

Interested in finding out more about our thin wall bolts? Get in touch and speak to a member of the BlindBolt team today on 012199 272 955 or send us an email on enquiries@blindbolt.com. We look forward to hearing from you.